

Module 5: Why do we need RNASeq?

February 09, 2026

Key concepts

- Quality control (continuation of Module 4)
- Gene-environment interactions
- Applications:
 - Alternative splicing events
 - Structural variant detection

Sequence Quality

Quality Score (Q) = $-10 \log_{10}(e)$, where **e** is the estimated probability of the base call being wrong.

- **Higher Q scores** → Smaller probability of error
- **Lower Q scores** → Higher probability of error, leading to:
 - Unusable reads
 - False-positive variant
 - calls

Inaccurate conclusions

Quality Score Thresholds

- **Q20**: Error rate=1%; Call accuracy = **99%** **Q30**: Virtually all
- reads are error-free (**benchmark for NGS**)

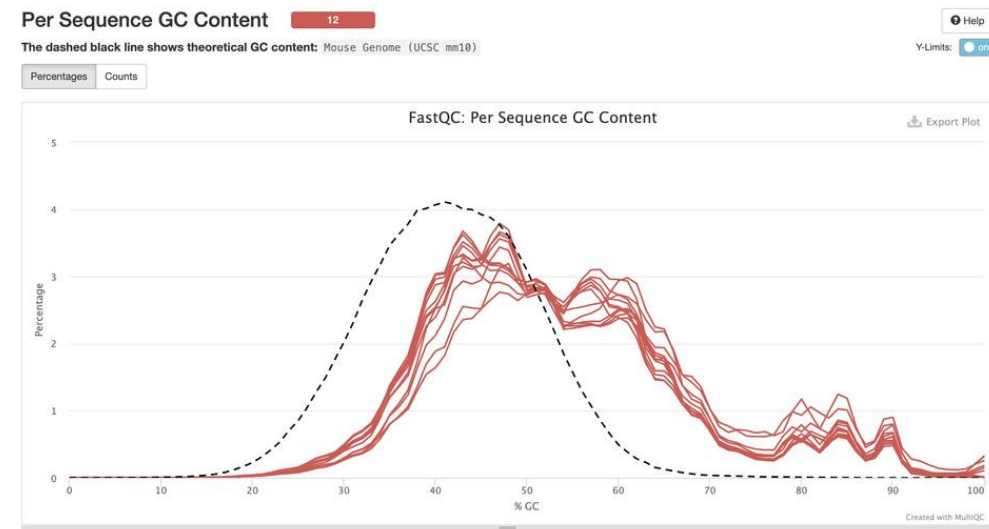
GC Content

- The GC content of the reads should be similar to the expected % GC for the organism (see reference table [here](#)).

The distribution should be normal

- unless over-represented sequences (sharp peaks on a normal distribution) or contamination with another organism (broad peak).

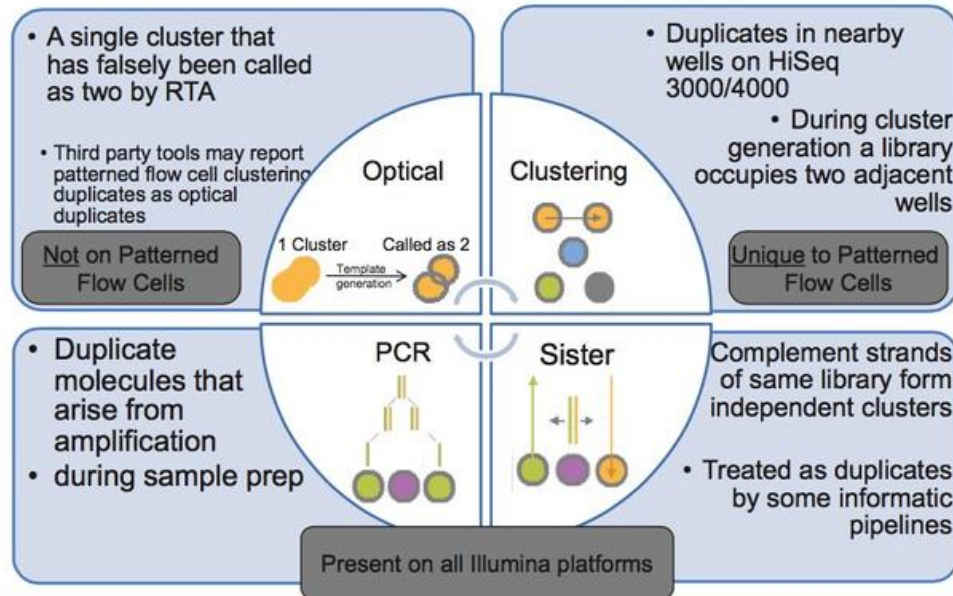
What happens when you align the reads to the wrong reference genome?



Overrepresented sequences (K-mers)

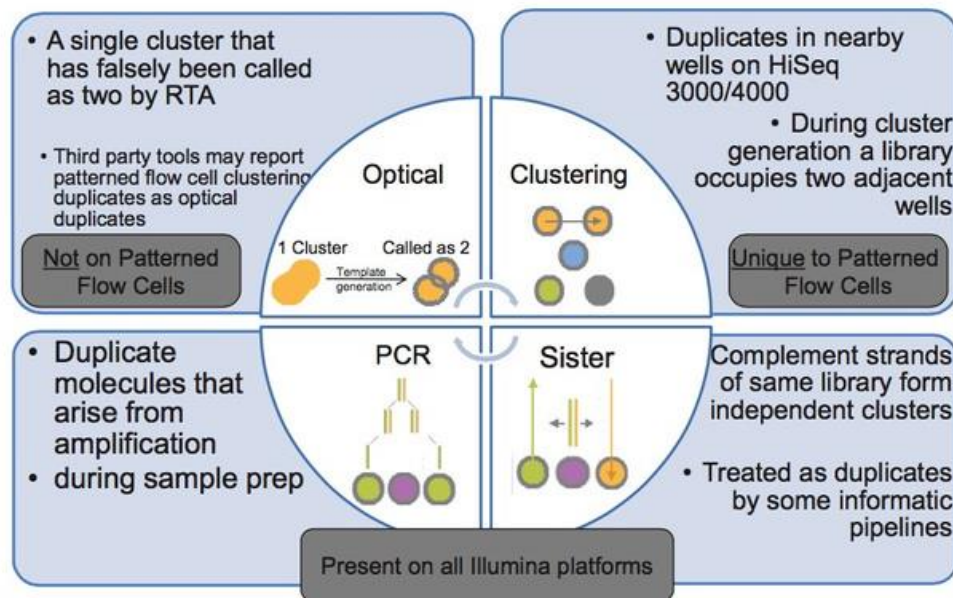
- The sequences (at least 20 bp) that occur in more than 0.1% of the total number of sequences
- Overrepresented sequences are usually adapters from the library prep and requires adapter trimming using tools like fastp or trimgalore

Read Duplication



- **Optical duplicate:** a duplicate read that occurs when a sequencer separates a single cluster into multiple clusters.
- Duplicates unique to patterned flow cells arise from when the library molecules from a cluster return to the surrounding solution and then act as a seed for second cluster in a nearby flow cell.

Read Duplication



- A **PCR duplicate** is a duplicate read that arises from occur in library preparation during the PCR amplification stage.
- Sister duplicates arise when the same DNA strand is used to form multiple clusters on the flow cell.

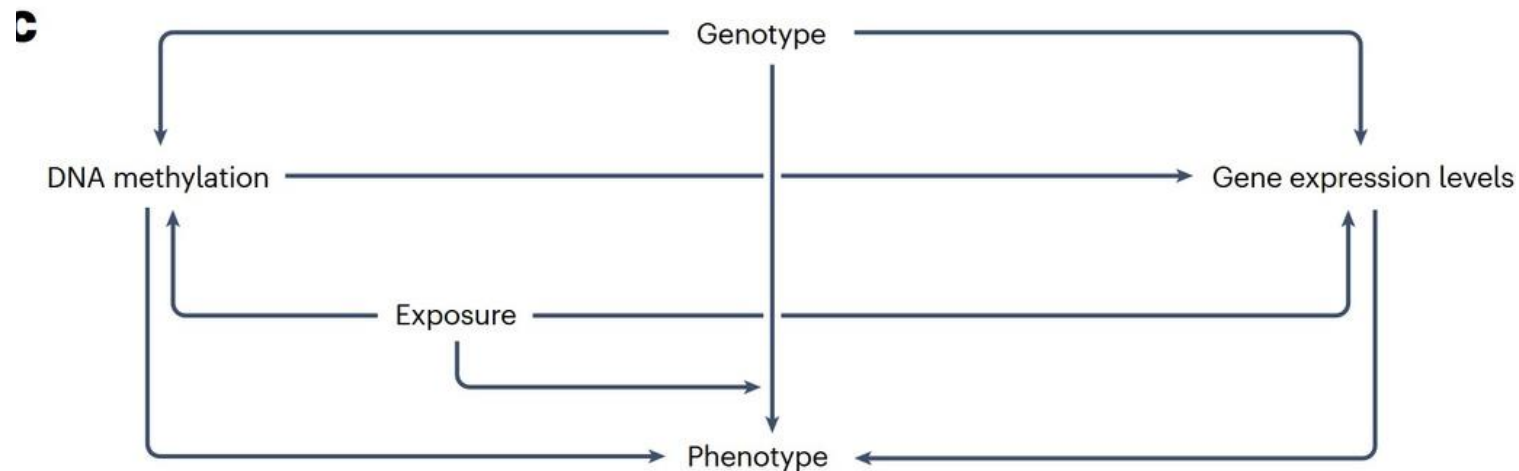
Read Duplication

- **Single-end** reads tend to have higher duplication rate than **paired-end** reads.
- **Higher** sequencing depth usually lead to **higher** duplication rate. **High** duplication
- rate does not necessarily indicate problems with the library preparation.
- **Highly** expressed genes have many copies of mRNA molecules, thus are expected to have **many** duplicate reads in the library, while **lowly** expressed genes should not.

Genotype × environment interactions (GxE)

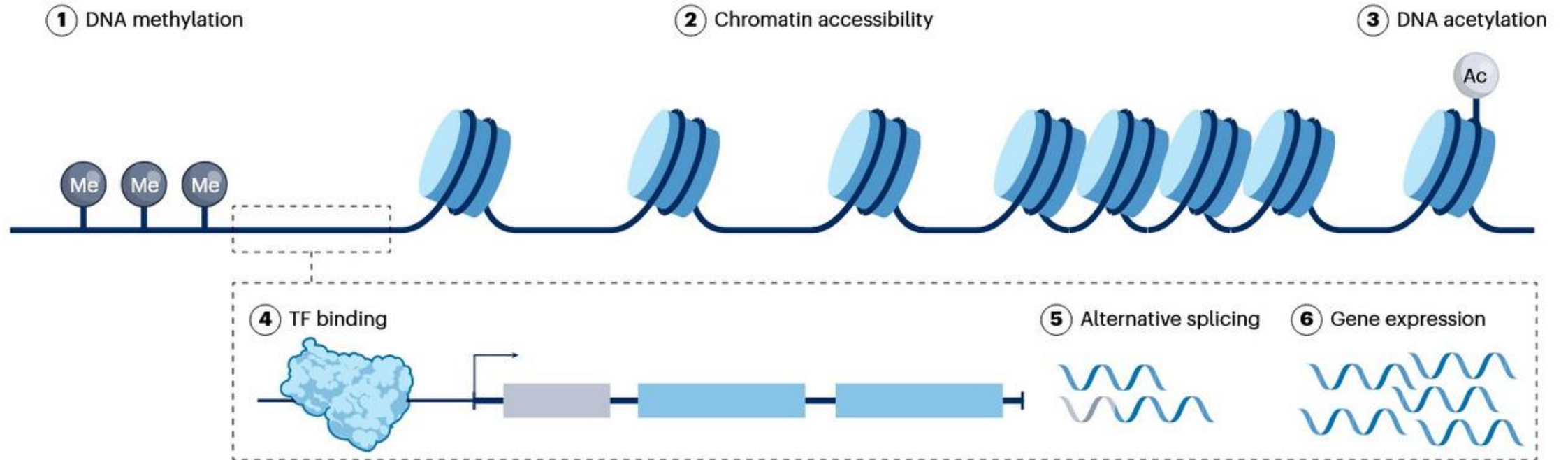
GxE occur when the genetic effect on a phenotype varies depending on the environment or, vice versa, the effect of the environment on a phenotype varies depending on the individual's genotype.

- Does a polygenic G × E interaction contribute to a human trait (and by how much) across one or more environmental contexts?

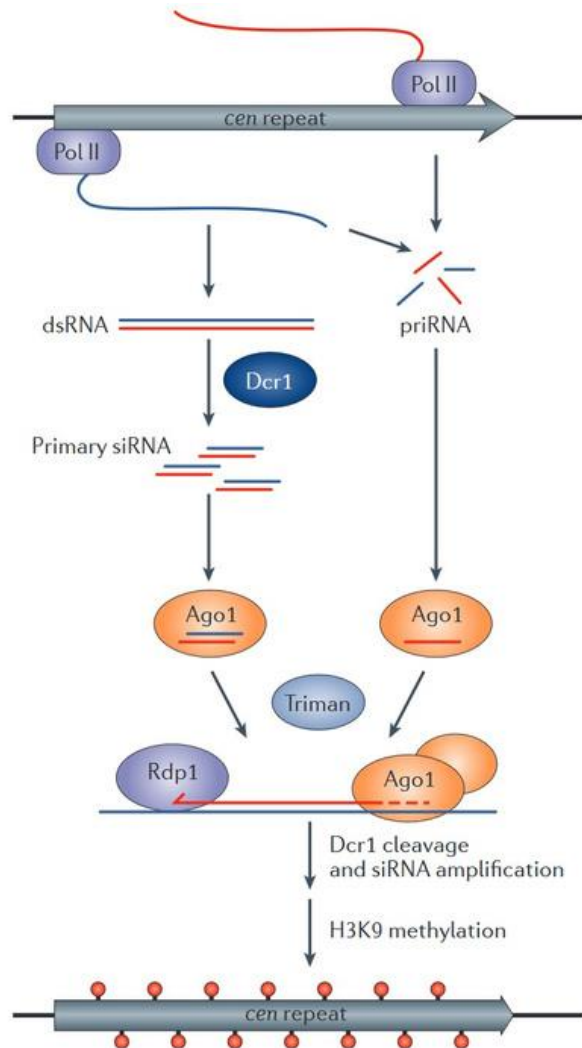


GxE interactions

Different gene-regulatory mechanisms can be impacted by GxE.



Epigenetics in complex diseases



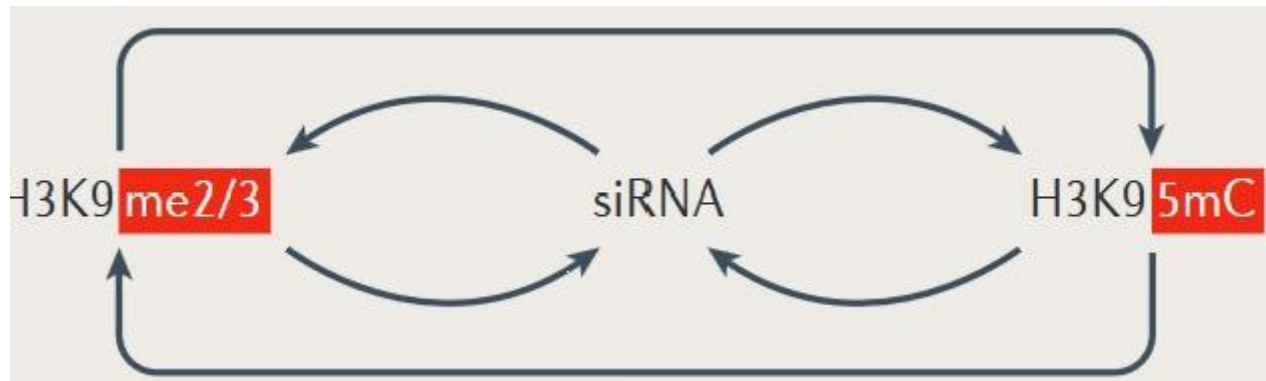
RNA interference (RNAi): various RNA silencing pathways that use small RNAs, together with a member of the conserved Argonaute (AGO) and PIWI family of proteins, to target genes for inactivation at the post-transcriptional or transcriptional levels

- *Drosophila melanogaster* and mammals have small RNAs--PIWI-interacting RNAs (piRNAs)-- which mediate RNA degradation in the cytoplasm and DNA or histone methylation in the nucleus

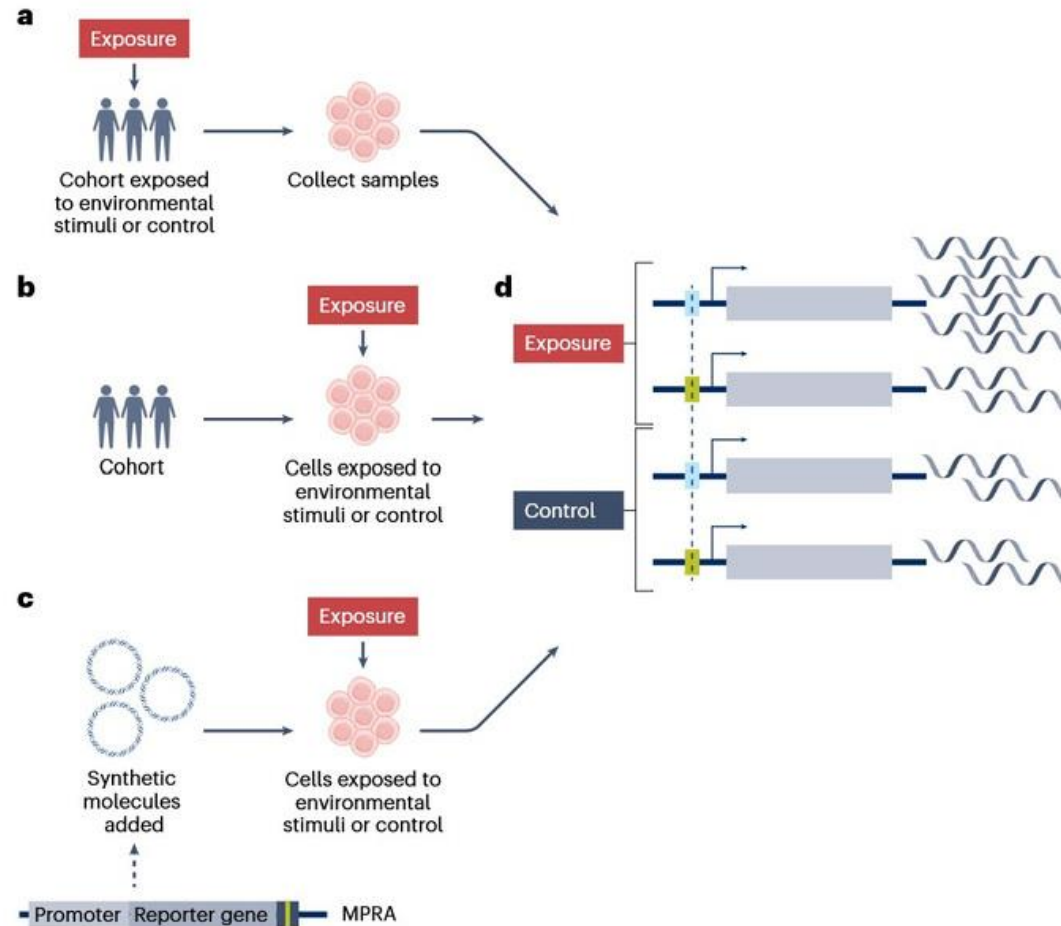
Epigenetics in complex diseases

Self-reinforcing positive feedback loops are formed by the functional coupling of different types of signal generation events.

- You can understand the molecular basis of these coupling events by identifying the proteins that recognize either histone H3K9me2 and H3K9me3, or DNA cytosine methylation (DNA-5mC), and the proteins that recruit enzymes (e.g. siRNAs) which catalyse the other methylation events.



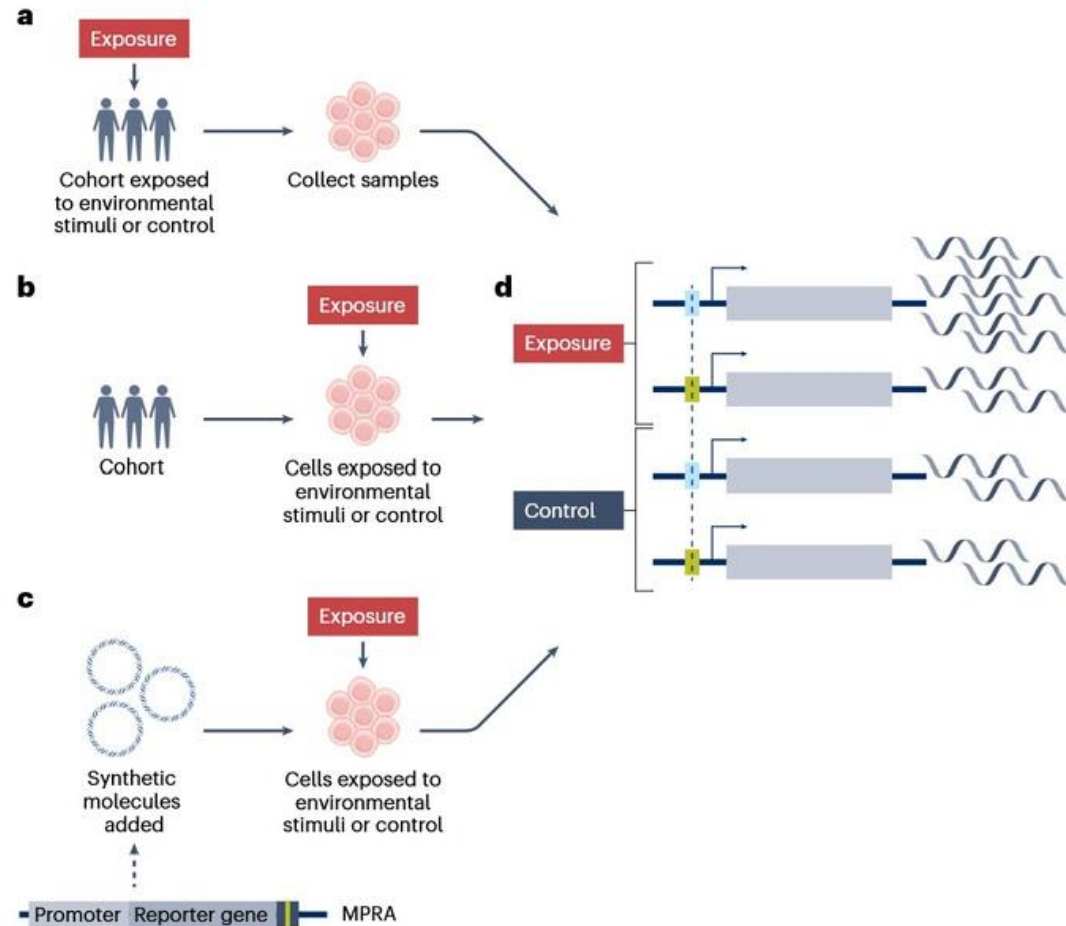
GxE interactions: Study Designs



A. The environment is measured in a cohort, biological samples are collected, and both molecular readouts (e.g., gene expression) and genotypes are assessed.

B. Biological samples are collected from a cohort, then deliberately exposed to an environment; afterwards, molecular readouts and genotypes are measured.

GxE interactions: Study Designs



C. Engineered constructs (carrying a minimal promoter, a reporter gene, and a sequence of interest) are introduced into cells, and molecular readouts are compared across different alleles and environmental conditions.

Example: Molecular GxE Studies using Immune Stimuli

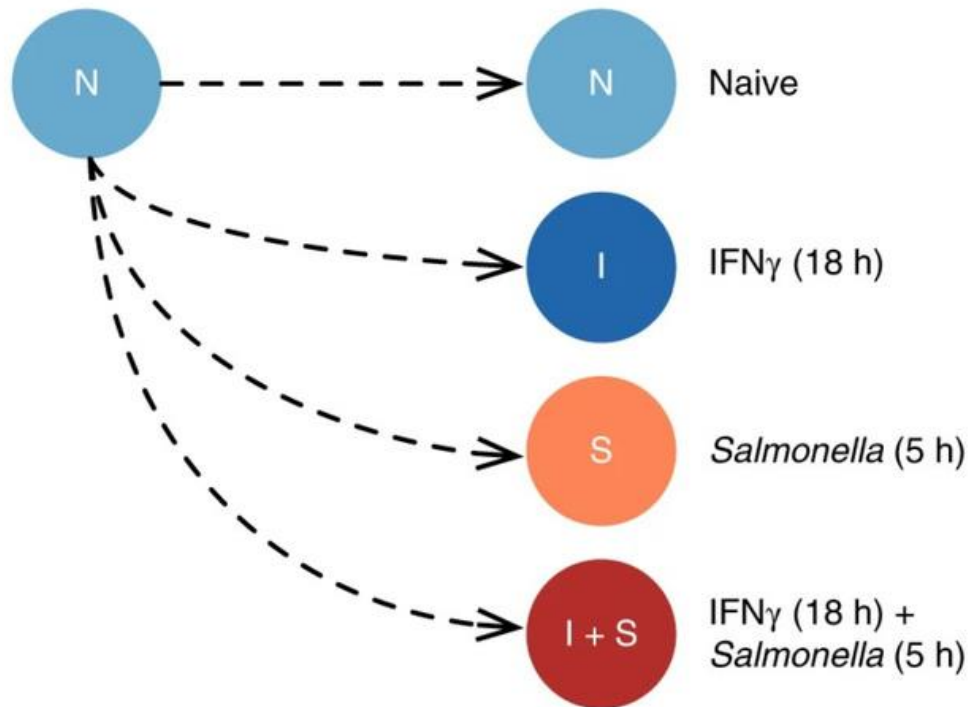
Experimental objectives:

- Understanding molecular mechanisms such as binding of transcription factors that are activated in response to immune stimuli.
- Find complex interactions, such as GxE with sex or age, duration of the exposure or cell type
- Determine interindividual differences in pathogen response and autoimmune disorders

Example: Molecular GxE for environmental exposures

Experimental design

Macrophages



Sample sizes

RNA

ATAC

84

42

84

41

84

31

84

31

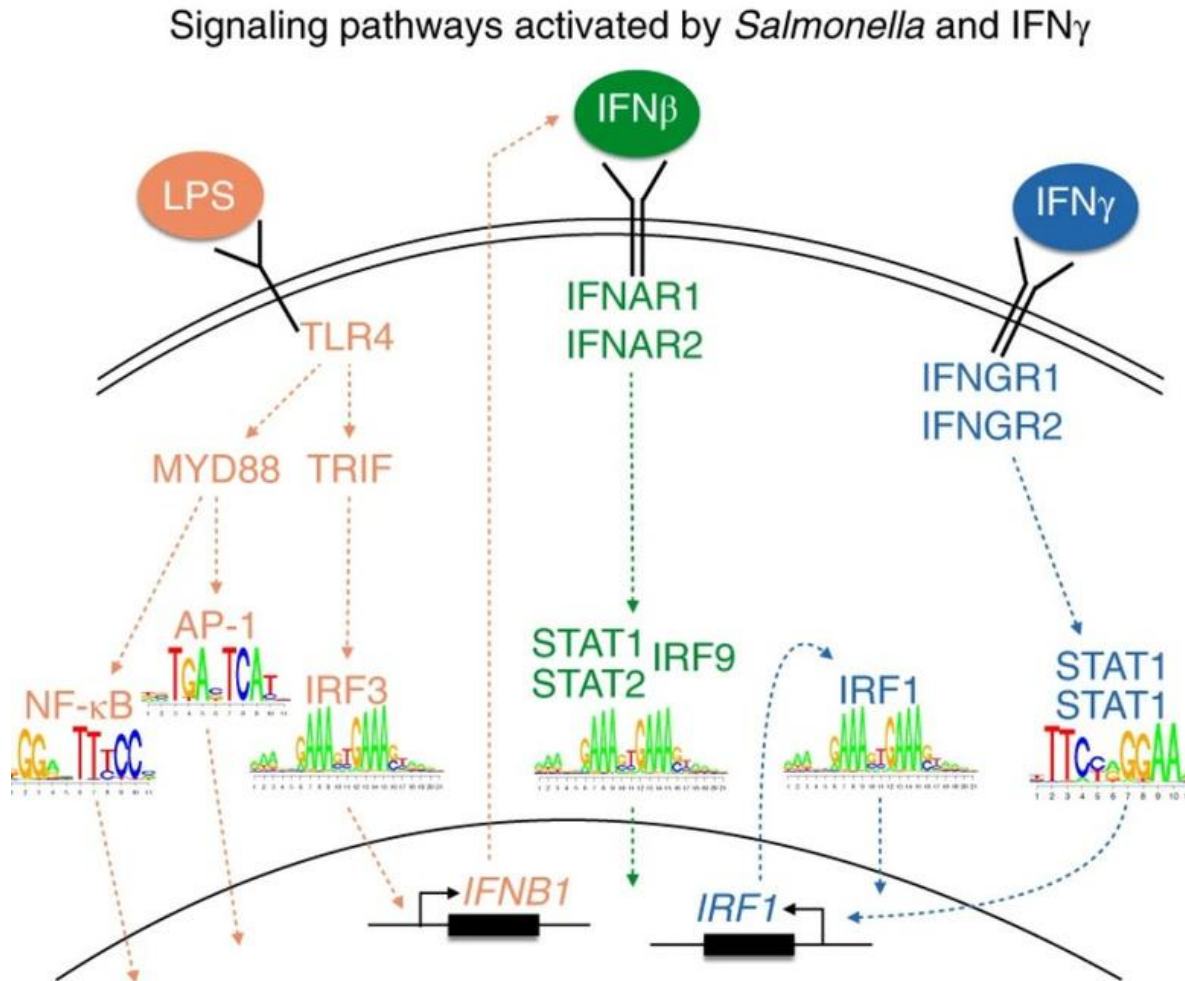
What are the molecular mechanisms for enhancer priming in immune response?

- Used ATAC-Seq and RNA-Seq to measure chromatin

accessibility and gene expression in human

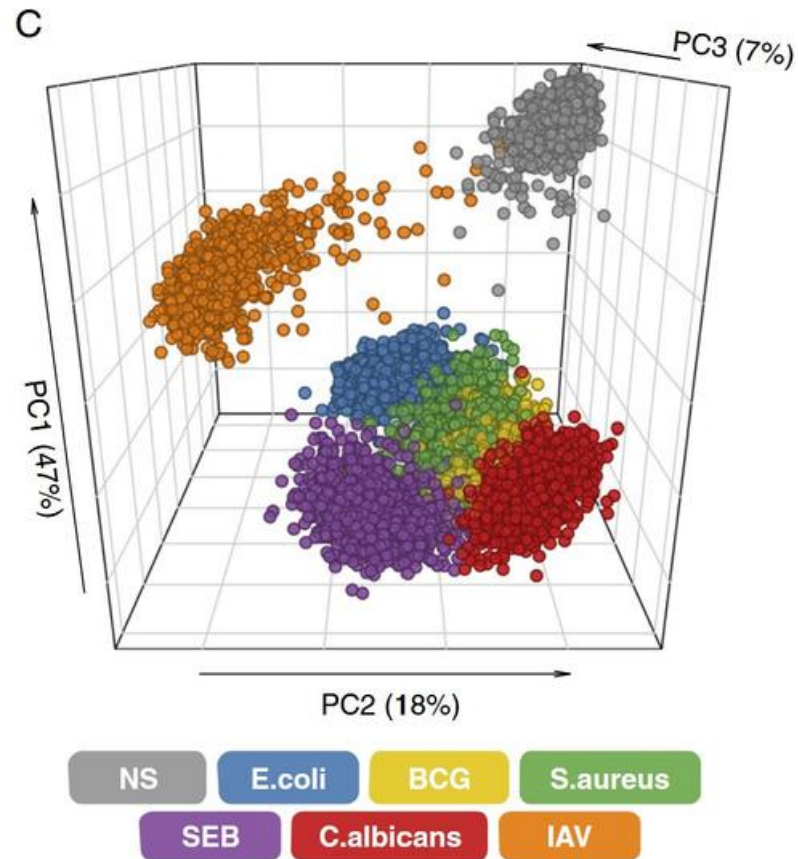
macrophages exposed to IFN γ , Salmonella and IFN γ plus Salmonella

Example: Molecular GxE for environmental exposures



Identified variants influencing the binding of cell-type-specific transcription factors, such as PU.1, which can then indirectly alter the binding of stimulus-specific transcription factors, such as NF- κ B or STAT2.

Example: Molecular GxE Studies using Immune Stimuli



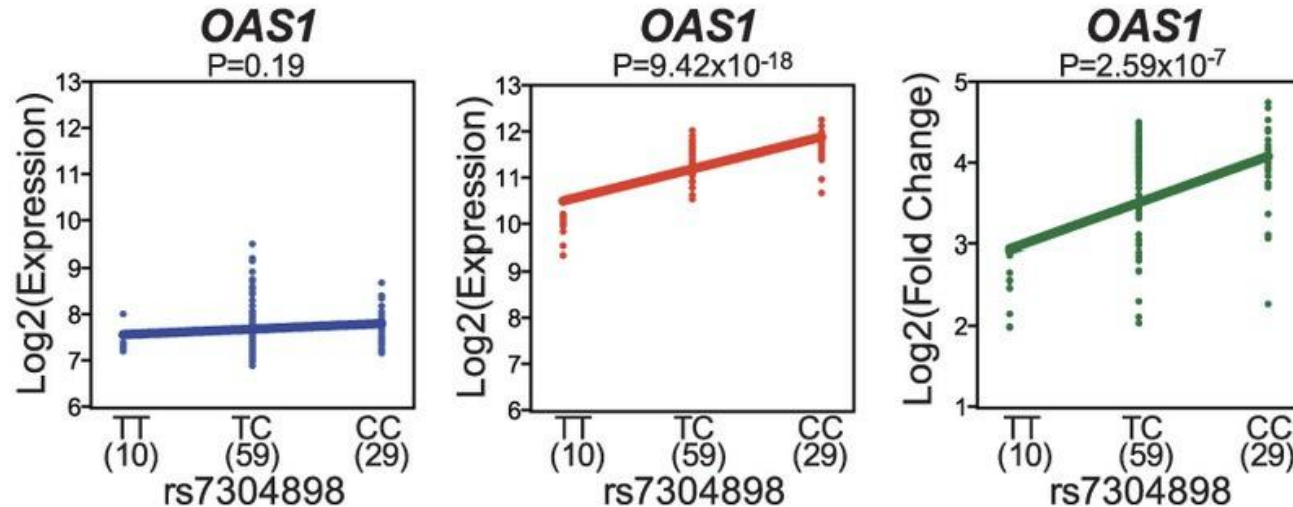
Do age and sex influence the gene expression of immune-related genes upon immune stimuli?

- Stimulated whole blood samples with three bacteria (*E. coli*, *S. aureus*, and *B. Calmette–Guérin*); a fungus (*C. albicans*); a live virus, *influenza A virus* (IAV); and a superantigen, *staphylococcal enterotoxin B* (SEB)

Example: Molecular GxE Studies using Immune Stimuli

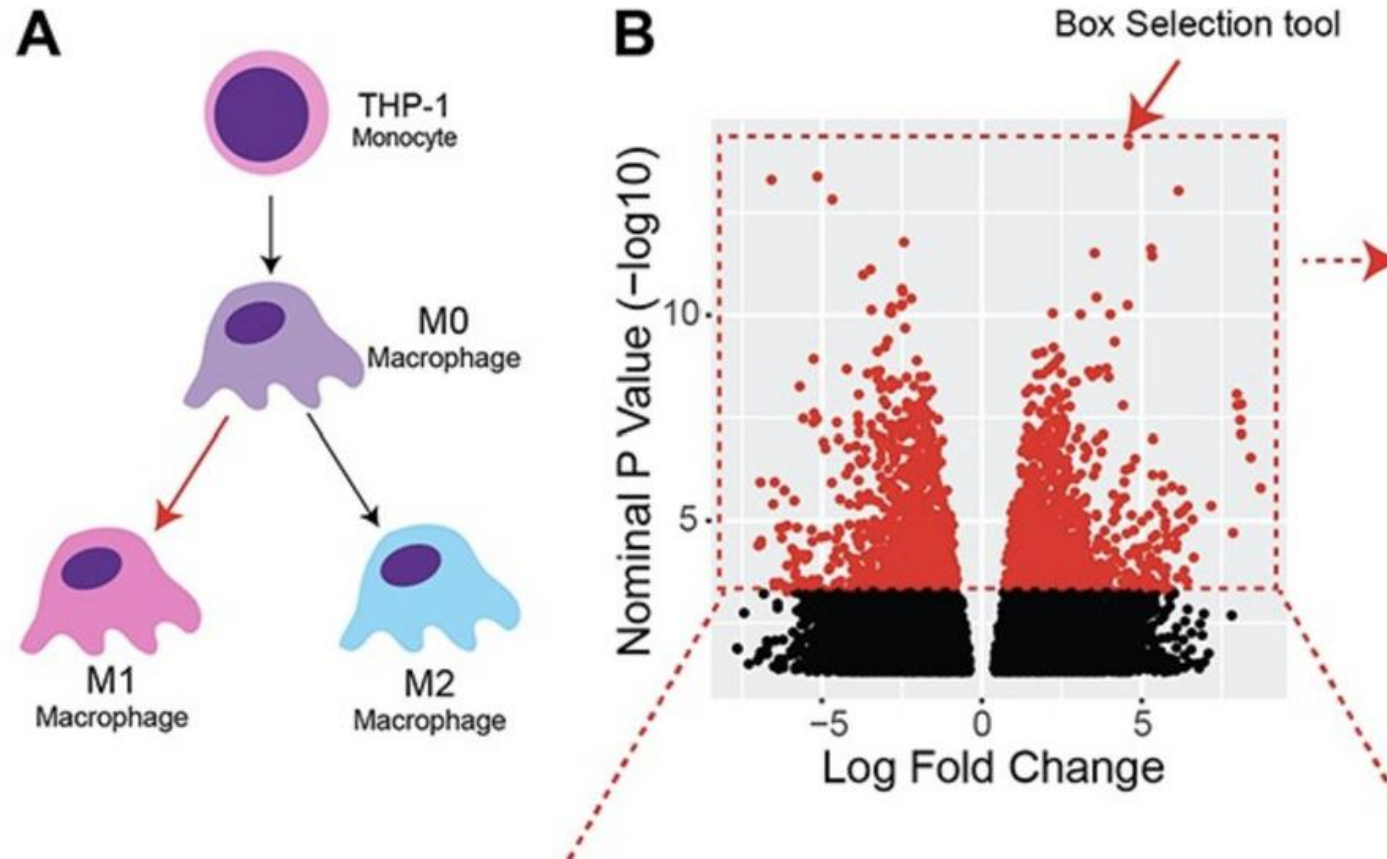
Is host genetic variation associated with variation in gene expression response to RV infections between individuals?

- OSA1 is a gene with known functions in viral response

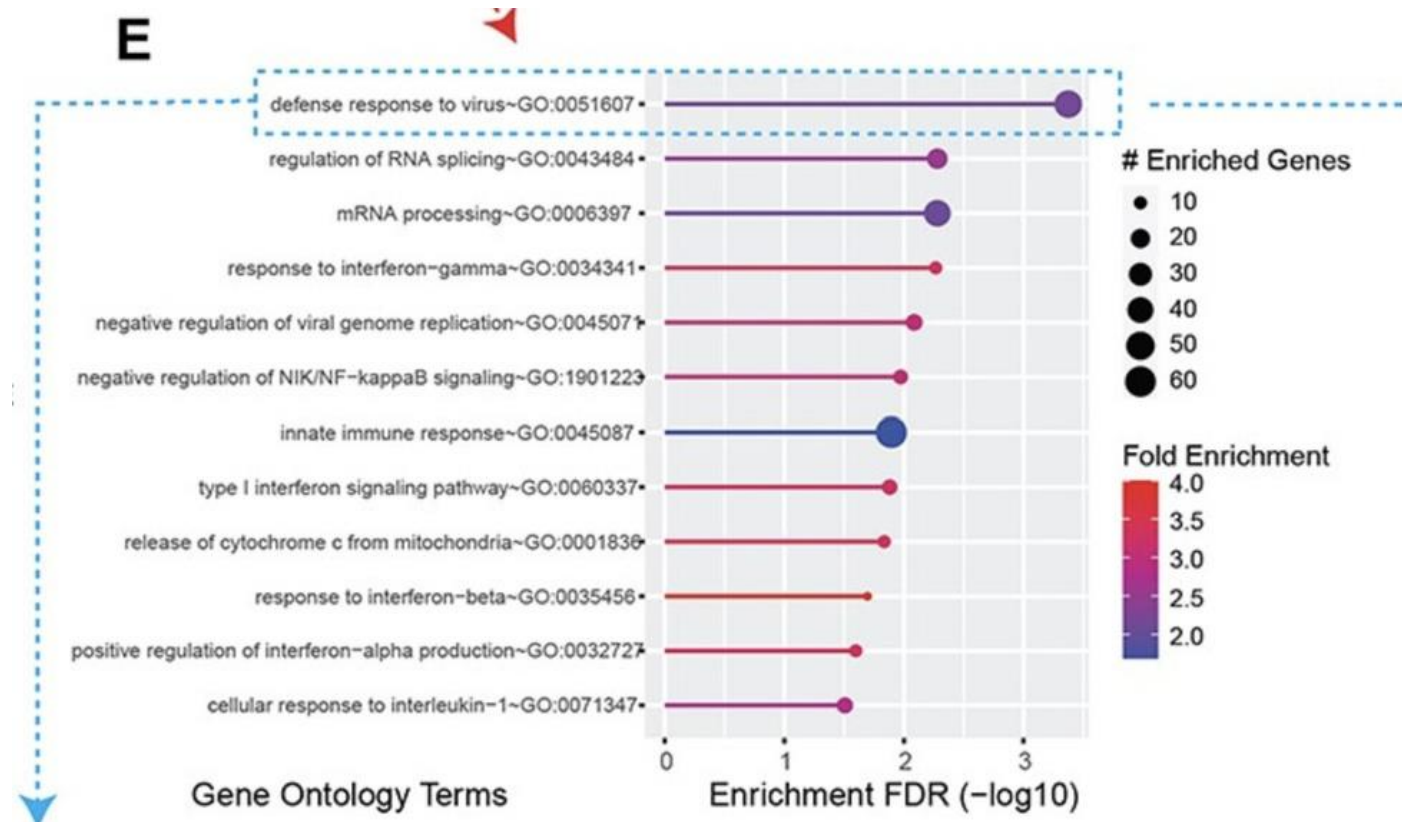


- uninfected cells (blue) RV-
- infected cells (red) reQTLs
- (green): differences in gene expression response

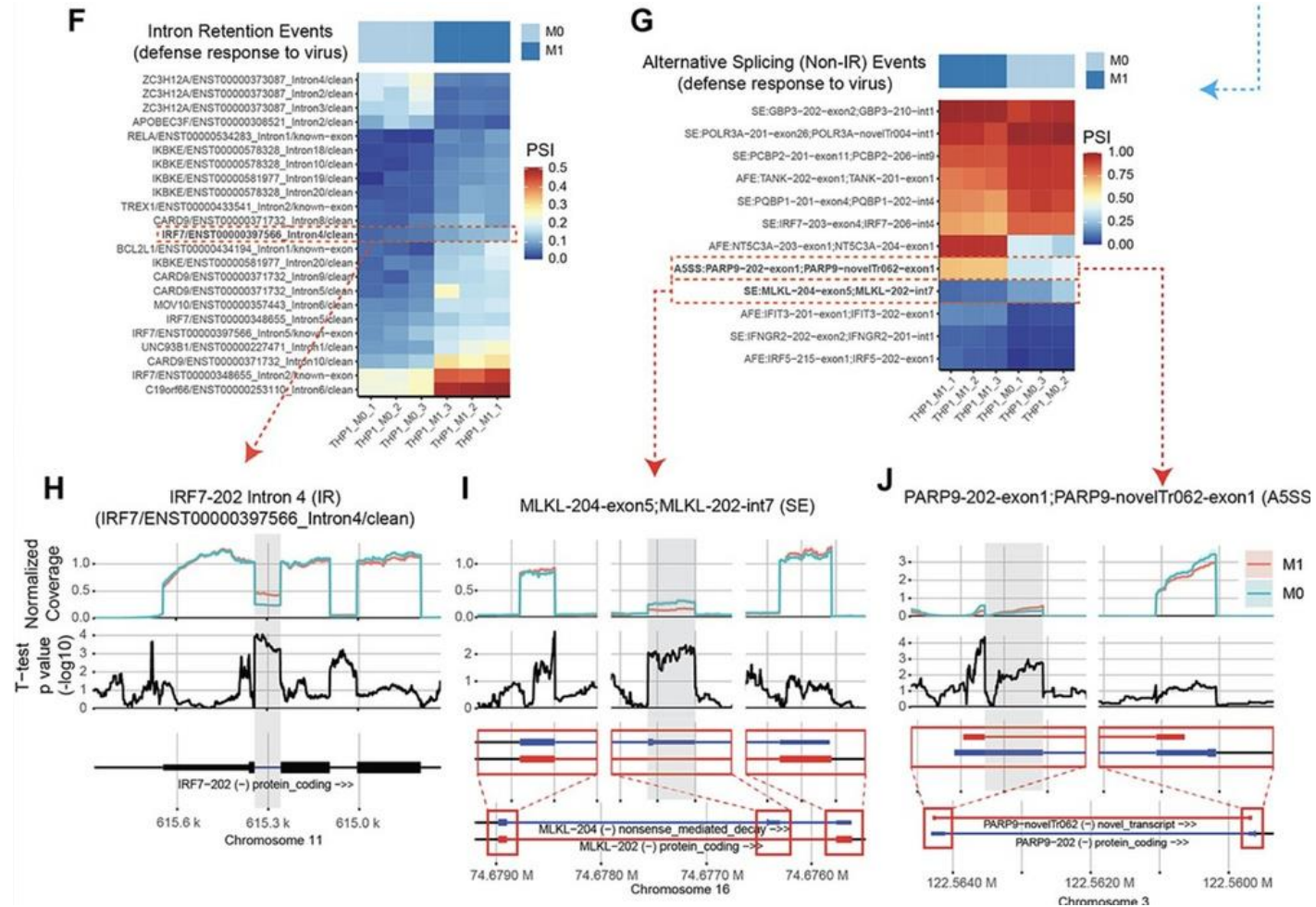
Applications: Alternative splicing events



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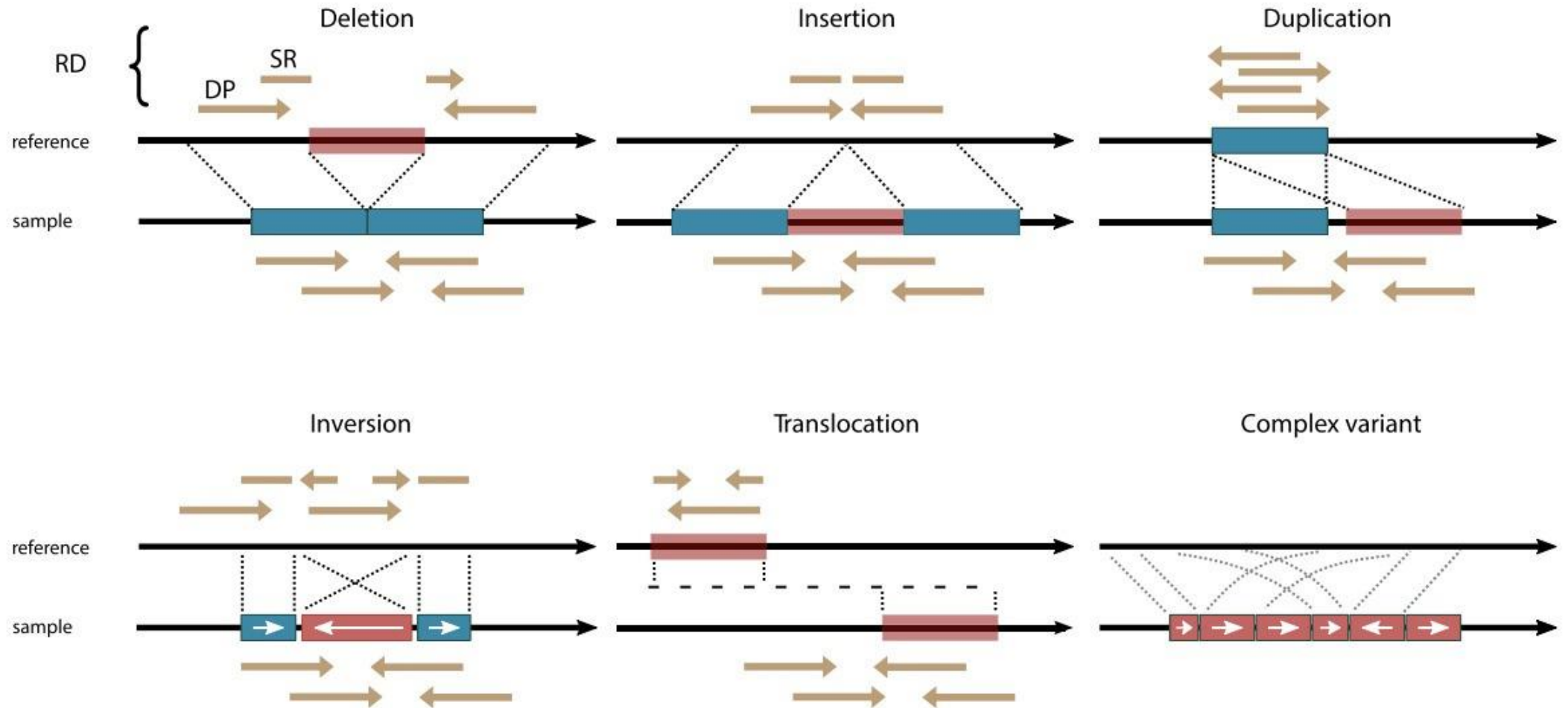


Applications: Alternative splicing events



Applications: Structural variant detection

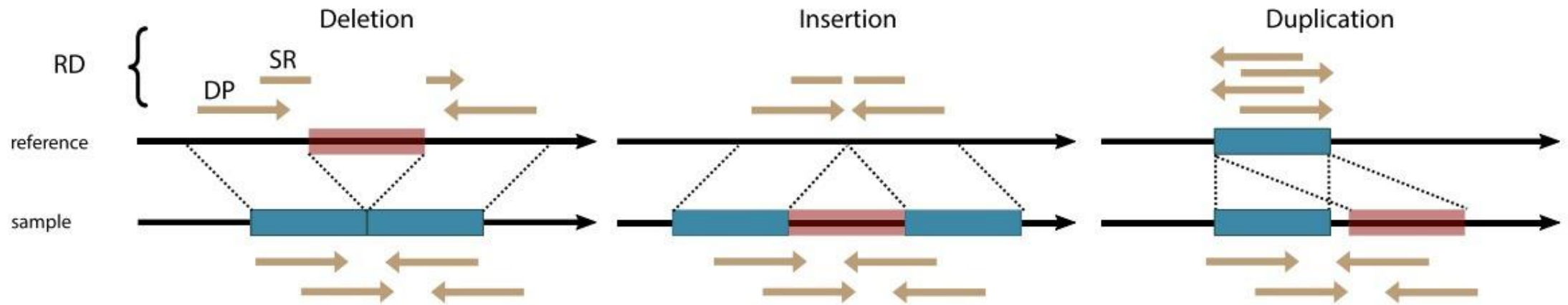
- SVs are genomic alterations **larger than 50 base pairs**



Applications: Structural variant detection

- SVs often indicate disruptions in genome stability Useful for identifying structural
- changes in genetic disorders and understanding complex genome rearrangements
- Cancer genomics and evolution

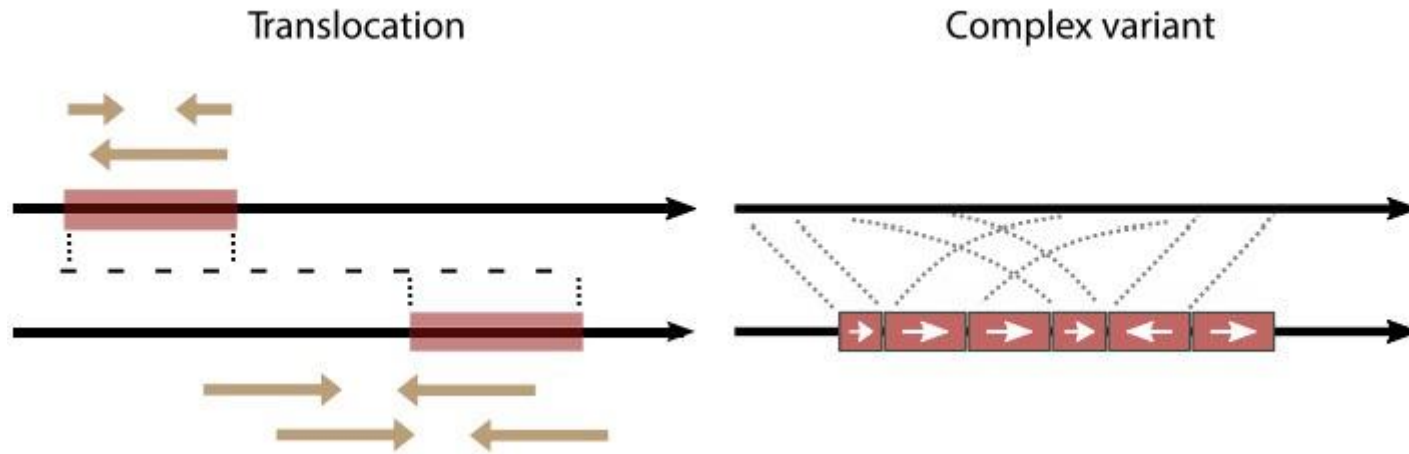
Applications: Structural variant detection (Patterns to look for)



1. Read Depth (RD):

- Changes in sequencing depth or coverage highlight larger duplications or deletions.
- Primarily used to detect **Copy Number Variants (CNVs)**.

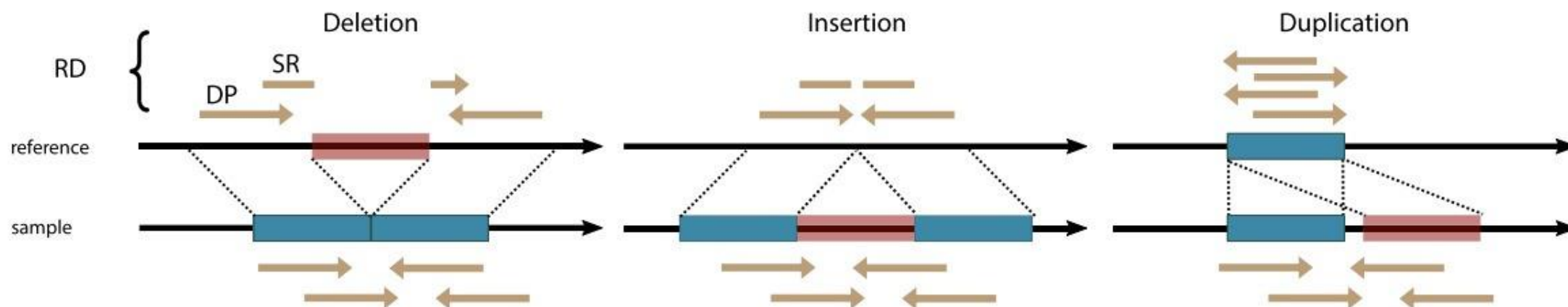
Applications: Structural variant detection (Patterns to look for)



2. Discordant Pairs (DP):

- Paired-end reads align to the reference at unexpected distances or orientations.
- Best suited for detecting **large SVs** such as inter-chromosomal **translocations** and **inversions**.

Applications: Structural variant detection (Patterns to look for)



3. Split Reads (SR):

- Reads span breakpoint regions and align partially to reference sequences.
- Detects **small SVs** with **base-pair resolution**.